



JOBSHEET 4

SELECTION 1

1. Objectives

1. Students are able to solve problems/case studies using simple selection syntax
2. Students are able to apply simple selection syntax in a Java program

2. Lab Work

2.1 Experiment 1: Using IF and IF-ELSE to Print the KRS

Experiment Time: 40 minutes

At the beginning of every semester, students must print their KRS (Study Plan Card) so it can be signed by their Academic Advisor (DPA). SIAKAD will check the student's UKT (tuition fee) payment status. If the student has fully paid the UKT, the system shows the KRS so it can be printed. Based on this case, create a Java program with the following steps.

1. Create a new file named **SelectionIfAttendanceNo.java**
2. Create the basic structure of a Java program with the **main()** function
3. Import the Scanner library and declare a scanner named **sc**
`Scanner sc = new Scanner(System.in);`
4. Add the following code to read input from the keyboard and store it in the **uktPaid** variable of type boolean

```
System.out.println(x: "--- Print KRS SIAKAD ---");  
System.out.print(s: "Has the UKT been paid? (true/false): ");  
boolean uktPaid = sc.nextBoolean();
```

5. Create an IF selection structure to check whether the UKT has been paid

```
if (uktPaid) {  
    System.out.println(x: "UKT payment verified");  
    System.out.println(x: "Please print your KRS and ask your DPA to sign it");  
}
```

6. Compile and run the program so that it shows the following output.

```
--- Print KRS SIAKAD ---  
Has the UKT been paid? (true/false): true  
UKT payment verified  
Please print your KRS and ask your DPA to sign it
```

Questions

1. What value must you enter so that both lines inside the IF block are printed? Explain why only that value is accepted!
2. Run the program, then enter **false**. Which lines are printed and which lines are not? Explain the execution flow when the IF condition is false!
3. Run the program, then enter **TRUE** (in capital letters) and **yes**. What happens with each input? If the program stops with an error, explain the cause!



4. The system needs to give information when the user enters the value false, with the output "Registration rejected. Please pay your UKT first". Modify the program by adding an ELSE structure, then show the run results for the inputs **true** and **false**!

2.2. Experiment 2: SWITCH-CASE to Print the KRS

Experiment Time: 60 minutes

At the beginning of every semester, students must print their KRS so it can be signed by their Academic Advisor (DPA). The SIAKAD system will check the student's current semester, then show the KRS for that semester so it can be printed. Based on this case, create a Java program with the following steps.

1. Create a new file named **SelectionSwitchAttendanceNo.java**
2. Create the basic structure of a Java program with the **main()** function
3. Import the Scanner library and declare a scanner named **sc**
4. Add the following code to read input from the keyboard and store it in the **semester** variable of type int

```
System.out.println(x: "--- Print KRS SIAKAD ---");  
System.out.print(s: "Enter your current semester: ");  
int semester = sc.nextInt();
```

5. Create a SWITCH-CASE selection structure to check the current semester

```
switch (semester) {  
    case 1:  
        System.out.println(x: "KRS for Semester 1 is displayed");  
        break;  
    case 2:  
        System.out.println(x: "KRS for Semester 2 is displayed");  
        break;  
    case 3:  
        System.out.println(x: "KRS for Semester 3 is displayed");  
        break;  
    case 4:  
        System.out.println(x: "KRS for Semester 4 is displayed");  
        break;  
    case 5:  
        System.out.println(x: "KRS for Semester 5 is displayed");  
        break;  
    case 6:  
        System.out.println(x: "KRS for Semester 6 is displayed");  
        break;  
    case 7:  
        System.out.println(x: "KRS for Semester 7 is displayed");  
        break;  
    case 8:  
        System.out.println(x: "KRS for Semester 8 is displayed");  
        break;  
    default:  
        System.out.println(x: "Invalid semester");  
}
```

6. Compile and run the program so that it shows the following output.



```
--- Print KRS SIAKAD ---  
Enter your current semester: 5  
KRS for Semester 5 is displayed
```

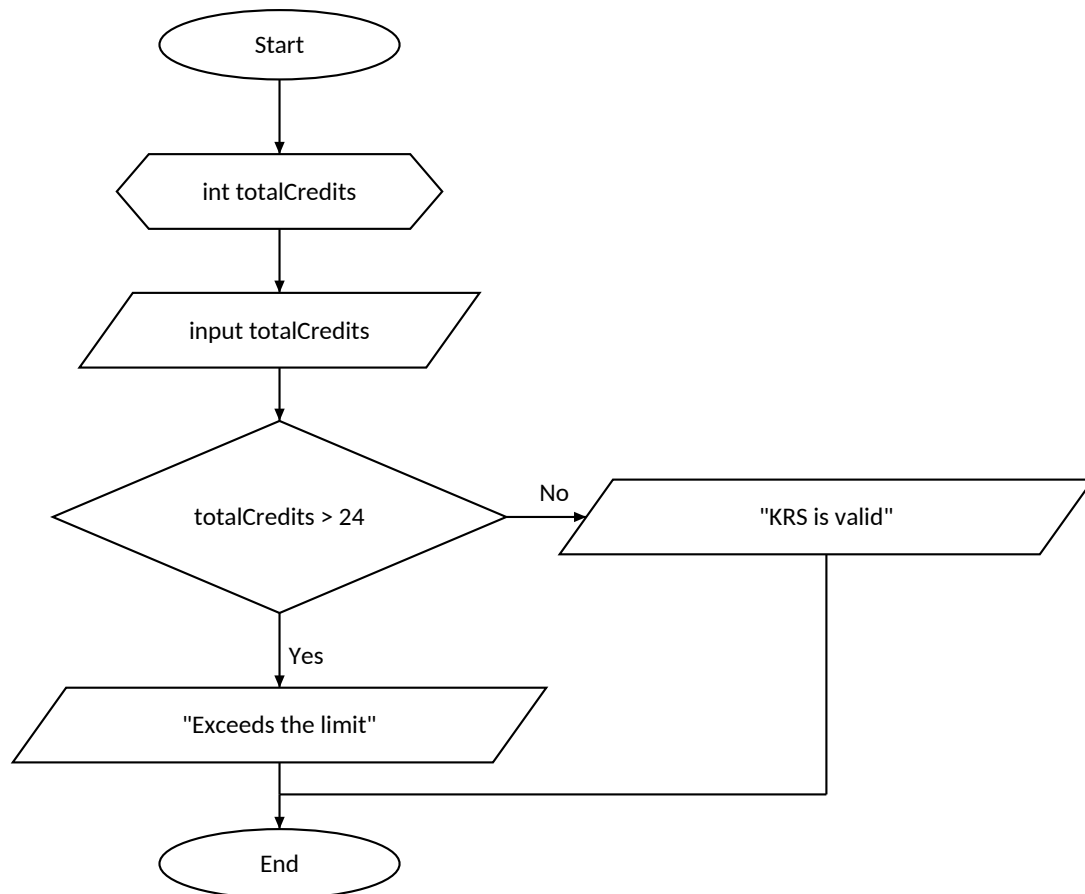
Questions

1. Delete the **break;** statement in **case 5**, then compile and run the program again with the input **5**. Write down the output, then explain the function of break in the SWITCH-CASE structure based on your experiment! Put the code back to how it was when you are done.
2. Run the program with the input **10**, then with the input **0**. What is the output of these two runs? Based on the results, explain the role of **default** and what will happen to the program if the default part is deleted!
3. Change the data type of the **semester** variable to **double**, then compile the program. Does the program compile successfully? Write down the error message and explain its cause. List the data types that can be used as the expression in a switch!
4. Create a new file named **SelectionIfElseAttendanceNo.java**. Convert the KRS printing program that uses SWITCH-CASE into an IF - ELSE IF - ELSE form. The program output must be exactly the same as the SWITCH-CASE version, including for invalid input. In your opinion, which one is easier to read for this case, and why?

3. Assignments

Assignment Time: 120 minutes

1. Open the file **SelectionIfAttendanceNo.java** again. Change the **IF-ELSE** selection structure in the program into a **Ternary Operator**, with the following rules:
 - a. The decision result is first stored in a String variable named **message**, then printed using a single `System.out.println()` statement
 - b. The program output must be exactly the same as the original program
 - c. Save it with the file name **Assignment1SelectionAttendanceNo.java**In your opinion, when is the Ternary Operator better to use than IF-ELSE, and when should it not be used?
2. Look at the following flowchart.



A KRS system validates the number of credits (SKS) taken by a student, where the maximum number allowed is 24 credits. Implement the flowchart above as a Java program using an IF-ELSE selection structure, then save it with the file name **Assignment2SelectionAttendanceNo.java**!

3. In the Basic Programming class, you made flowcharts and pseudocode for two cases on the Exercise slides (page 33). Now implement both of them in Java, with the following rules.
 - a. For Problem 1 — Parking System, use an IF-ELSE selection structure and name the file **AssignmentParkingAttendanceNo.java**
 - b. For Problem 2 — Academic Queue Machine, use a SWITCH-CASE selection structure and name the file: **AssignmentQueueAttendanceNo.java**. In addition, the program must include a default part to handle codes outside 1–4 with the message "Service code is not available"